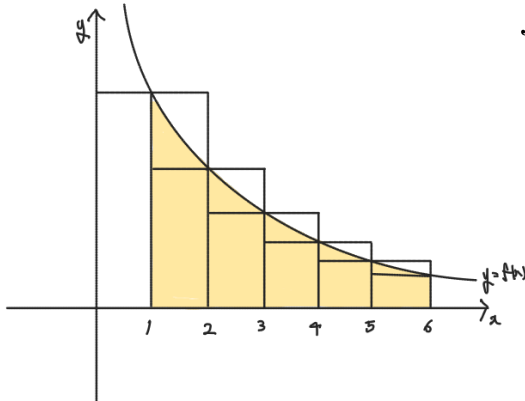


2026년 1학기 수학1 4차 손과제				
학과		학번		이름

§11.3 The Integral Test (적분판정법)

1. Suppose f is a continuous positive decreasing function for $x \geq 1$ and $a_n = f(n)$. By drawing a picture, rank the following three quantities in increasing order:

$$\int_1^6 f(x) dx, \quad \sum_{i=1}^5 a_i, \quad \sum_{i=2}^6 a_i$$



$$\sum_{i=2}^6 a_i < \int_1^6 f(x) dx < \sum_{i=1}^5 a_i$$

2. Determine whether the series is convergent or divergent.

(1) $\sum_{n=1}^{\infty} \frac{1}{(3n-1)^4} = a_n = f(n)$

$\frac{1}{(3n-1)^4}$ is decreasing, positive, continuous $\rightarrow f(x) = \frac{1}{(3x-1)^4}$ is continuous, positive & decreasing on $[1, \infty)$

By The Integral Test, if $\int_1^{\infty} \frac{1}{(3x-1)^4} dx$ is conv, the series is conv.

Set $3x-1=u$, $du=3 \cdot dx$ $\int_2^{\infty} \frac{1}{u^4} \cdot \frac{1}{3} du = \lim_{t \rightarrow \infty} \left[-\frac{1}{3} u^{-3} \right]_2^t = \frac{1}{8}$, conv. $\rightarrow \int_1^{\infty} f(x) dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{(3x-1)^4} dx = \lim_{t \rightarrow \infty} \left[-\frac{1}{9} (3x-1)^{-3} \right]_1^t$

Thus, the series is convergent.

$= \frac{1}{12}$, conv.

(2) $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^3}$

$\frac{1}{n(\ln n)^3}$ is decreasing, positive, continuous. $\rightarrow f(x) = \frac{1}{x(\ln x)^3}$ is continuous, positive & decreasing on $[2, \infty)$

By The Integral Test, if $\int_2^{\infty} \frac{1}{x(\ln x)^3} dx$ is conv, the series is conv.

Set $\ln x = u$, $du = \frac{1}{x} \cdot dx$ $\int_{\ln 2}^{\infty} \frac{1}{u^3} \cdot du = \lim_{t \rightarrow \infty} \left[-\frac{1}{2} u^{-2} \right]_{\ln 2}^t = \frac{1}{2(\ln 2)^2}$ $\rightarrow \int_2^{\infty} \frac{1}{x(\ln x)^3} dx = \lim_{t \rightarrow \infty} \int_2^t \frac{1}{x(\ln x)^3} dx$

Thus, the series is convergent.

$= \lim_{t \rightarrow \infty} \left[-\frac{1}{2(\ln x)^2} \right]_2^t = \lim_{t \rightarrow \infty} \left(\frac{1}{2(\ln 2)^2} - \frac{1}{2(\ln t)^2} \right)$
 $= \frac{1}{2(\ln 2)^2}$ conv.

$$(3) \sum_{n=2}^{\infty} \frac{1}{n \ln n}$$

$\frac{1}{n \ln n}$ is decreasing, positive, continuous. $\rightarrow f(x) = \frac{1}{x \ln x}$.
($n \geq 2$) ($n \geq 2$) ($n \geq 2$)

By The Integral Test, if $\int_2^{\infty} \frac{1}{x \ln x} \cdot dx$ is conv, the series is conv.

Set $\ln x = u$, $du = \frac{1}{x} \cdot dx$ $\int_{\ln 2}^{\infty} \frac{1}{u} \cdot du = \lim_{t \rightarrow \infty} [\ln |u|]_{\ln 2}^t = \infty$, div.

Thus, the series is divergent.

3.(1)[기출문제] For what values of p is the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ convergent? $p > 1$

(2)[기출문제] For what values of p is the series $\sum_{n=2}^{\infty} \frac{1}{n (\ln n)^{2p}}$ convergent? $p > \frac{1}{2}$
연속, 양수, 감소 $\rightarrow$ 적분정법 ($n \geq 2$)

(3) For what values of p is the series $\sum_{n=1}^{\infty} n(1+n^2)^p$ convergent? $p < -1$
($n \geq 2$)

(4) For what values of p is the series $\sum_{n=2}^{\infty} \frac{\ln n}{n^p}$ convergent? $p > 1$

§11.4 The Comparison Tests (비교판정법)

4. Determine whether the series converges or diverges.

$$(1) \sum_{n=1}^{\infty} \underbrace{\frac{n-1}{n^3+1}}_{a_n}$$

$0 \leq a_n < \frac{1}{n^2}$ for all n .

$\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent. (P-series with $p=2$)

By DCT, the series is convergent.

$$(2) \sum_{n=1}^{\infty} \underbrace{\frac{9^n}{3+10^n}}_{a_n}$$

$0 < a_n < \left(\frac{9}{10}\right)^n$ for all n .

$\sum_{n=1}^{\infty} \left(\frac{9}{10}\right)^n = \frac{\frac{9}{10}}{1 - \frac{9}{10}} = 9$, conv. $\sum_{n=1}^{\infty} \left(\frac{9}{10}\right)^n$ is a convergent geometric series with $r = \frac{9}{10}$.

By DCT, the series is convergent.

$$(3) \sum_{k=1}^{\infty} \underbrace{\frac{k \sin^2 k}{1+k^3}}_{a_n}$$

$$0 < a_n < \frac{1}{k^2} \text{ for all } n.$$

$$\sum_{k=1}^{\infty} \frac{1}{k^2} \text{ is convergent. (p-series with } p=2)$$

By DCT, the series is convergent.

$$(4) \sum_{n=1}^{\infty} \underbrace{\frac{1}{\sqrt{n^2+1}}}_{a_n}$$

$$b_n = \frac{1}{n}, \sum_{n=1}^{\infty} b_n \text{ is divergent. (p-series with } p=1)$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 1$$

By LCT, the series is divergent.

$$(5) \sum_{n=1}^{\infty} \underbrace{\sin\left(\frac{1}{n}\right)}_{a_n}$$

$$b_n = \frac{1}{n}, \sum_{n=1}^{\infty} b_n \text{ is divergent (p-series with } p=1)$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 1$$

By LCT, the series is divergent.

$$(6) \sum_{n=1}^{\infty} \underbrace{\sin^2\left(\frac{1}{n}\right)}_{a_n}$$

$$b_n = \frac{1}{n^2}, \sum_{n=1}^{\infty} b_n \text{ is convergent (p-series with } p=2)$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 1$$

By LCT, the series is convergent.

5. Prove that if $a_n \geq 0$ and $\sum a_n$ converges, then $\sum a_n^2$ also converges.

$\sum a_n$ 이 수렴하므로, $\lim_{n \rightarrow \infty} a_n = 0$ 이다.

충분히 큰 수 n 에 대하여 $a_n < 1$ 이다.

양변에 a_n 을 곱하면 $a_n^2 < a_n$ 이고,

$\sum a_n^2$ 은 수렴하므로 DCT에 따라서 $\sum a_n^2$ 은 수렴한다.

§11.5 Alternating Series and Absolute Convergence (교대급수와 절대수렴)

6. Test the series for convergence or divergence.

$$(1) -\frac{2}{5} + \frac{4}{6} - \frac{6}{7} + \frac{8}{8} - \frac{10}{9} + \dots$$

$$a_n = (-1)^n \underbrace{\frac{2n}{4+n}}_{b_n}$$

$\lim_{n \rightarrow \infty} b_n \neq 0 \Rightarrow \lim_{n \rightarrow \infty} a_n$ does not exist. $\therefore \sum a_n$ is divergent by the Test for Divergence.

~~By AST, the series is divergent.~~

$$(2) \frac{1}{\ln 3} - \frac{1}{\ln 4} + \frac{1}{\ln 5} - \frac{1}{\ln 6} + \frac{1}{\ln 7} - \dots$$

$$a_n = \frac{(-1)^{n+1}}{\ln(n+2)}, \quad b_n = \frac{1}{\ln(n+2)}$$

$$b_{n+1} \leq b_n \text{ for all } n$$

$$\lim_{n \rightarrow \infty} b_n = 0$$

By AST, the series is convergent.

$$(3) \sum_{n=1}^{\infty} (-1)^n \underbrace{e^{-n}}_{b_n}$$

$$b_{n+1} \leq b_n \text{ for all } n$$

$$\lim_{n \rightarrow \infty} b_n = 0$$

By AST, the series is convergent.

$$(4) \sum_{n=1}^{\infty} \frac{n \cos n\pi}{2^n} = (-1)^n$$

$$\frac{n}{2^n} = b_n$$

$$b_{n+1} \leq b_n \text{ for all } n \leftarrow \text{① } \frac{b_{n+1}}{b_n} = \frac{(n+1)}{2^{n+1}} \cdot \frac{2^n}{n} = \frac{n+1}{2n} \leq 1 \text{ for } n \geq 1. \text{ or } \text{② } f(x) = \frac{x}{2^x}, \quad f' < 0$$

$$\lim_{n \rightarrow \infty} b_n = 0$$

By AST, the series is convergent.

$$(5) \sum_{n=1}^{\infty} \underbrace{(-1)^n}_{a_n} \underbrace{\sin\left(\frac{\pi}{n}\right)}_{b_n}$$

$$n=1, a_n=0$$

$$b_{n+1} \leq b_n \text{ for } n \geq 2$$

$$\lim_{n \rightarrow \infty} b_n = 0$$

By AST, the series is convergent.

7. Determine whether the series is absolutely convergent, conditionally convergent or divergent.

$$(1) \sum_{n=1}^{\infty} \frac{\sin n}{2^n}$$

$$0 \leq \left| \frac{\sin n}{2^n} \right| \leq \frac{1}{2^n} \quad \& \quad \sum \left(\frac{1}{2}\right)^n \text{ is a convergent geometric series } \left(r = \frac{1}{2} < 1\right)$$

$$\sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{\frac{1}{2}} = 1, \text{ convergent.}$$

Thus, the series is absolutely convergent.

By DCT

$$(2) \sum_{n=1}^{\infty} \frac{\cos n\pi}{3n+2}$$

$$1) \frac{\cos n\pi}{3n+2} = \frac{(-1)^n}{3n+2} \quad b_n = \frac{1}{3n+2} \quad \begin{array}{l} \textcircled{1} b_{n+1} \leq b_n \\ \textcircled{2} \lim_{n \rightarrow \infty} b_n = 0 \end{array} \quad \text{By AST, the given series is convergent.}$$

$$2) \left| \frac{\cos n\pi}{3n+2} \right| = \frac{1}{3n+2} \quad b_n = \frac{1}{3n} \quad (\sum b_n \text{ diverges})$$

$$\lim_{n \rightarrow \infty} \frac{b_n}{a_n} = \lim_{n \rightarrow \infty} \frac{3n}{3n+2} = 1$$

By LCR, $\sum_{n=1}^{\infty} \left| \frac{\cos n\pi}{3n+2} \right|$ is divergent.

Thus, the given series is conditionally convergent.

8. Show that the series is convergent. How many terms of the series do we need to add in order to find the sum to the indicated accuracy?

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^6} \quad (|\text{error}| < \underline{0.00005})$$

$$b_n = \frac{1}{n^6}, \quad b_{n+1} \leq b_n \text{ for all } n$$

$$\lim_{n \rightarrow \infty} b_n = 0$$

By AST, the series is convergent.

$$|R_n| = |S - S_n| \leq b_{n+1} \quad \frac{1}{2^6} = \frac{1}{64}$$

$$\frac{1}{6^6} = \frac{1}{46656} < 0.00005, \quad n+1=6, \quad n=5$$

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↓

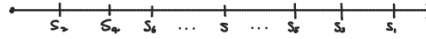
→ 교대급수

9. Is the 50th partial sum s_{50} of the alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ an overestimate or an underestimate of the total sum? Explain.

$$\left[\frac{(-1)^{n-1}}{n} \right] = \left[1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \dots \right]$$

$[S_{2n}]$: 단조증가, 유계 ($< S$)
(만조급수 수열)

$$\rightarrow \lim_{n \rightarrow \infty} S_{2n} = S$$



$\underbrace{\quad}_{\text{Underestimate}}$

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{49} - \frac{1}{50} + \dots$$

$$+ \left(\frac{1}{51} - \frac{1}{52} \right) + \left(\frac{1}{53} - \frac{1}{54} \right) + \dots$$

10. Find the values of p for which the series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^p}$ is convergent.

교대급수 판정법 (AST): ① $b_{n+1} \leq b_n$

$$\textcircled{2} \lim_{n \rightarrow \infty} b_n = 0$$

→ 급수 수렴

$$b_n = \frac{1}{n^p}$$

$$1) p > 0$$

$$\frac{1}{(n+1)^p} \leq \frac{1}{n^p}$$

$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$$

AST에 따라서 급수 수렴

$$2) p = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{n^0} = 1$$

한산 판정법에 따라서 급수 발산

$$3) p < 0$$

$$(n+1)^{-p} \geq n^{-p}$$

$$\lim_{n \rightarrow \infty} n^{-p} = \infty$$

한산 판정법에 따라서 급수 발산

$\underbrace{\quad}_{p > 0, n}$

§11.6 The Ratio Test (비 판정법)

11. Use the Ratio Test to determine whether the series is convergent or divergent.

$$(1) \sum_{n=1}^{\infty} \frac{10^n}{(n+1)4^{2n+1}} \quad \rightarrow a_n$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{10^{n+1}}{(n+2)4^{2n+3}} \times \frac{(n+1)4^{2n+1}}{10^n} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{5(n+1)}{8(n+2)} \right| = \frac{5}{8} < 1$$

By the Ratio Test, the given series is absolutely convergent. $\therefore$ converges //

$$(2) \sum_{n=1}^{\infty} \frac{\cos(n\pi/3)}{n!} \quad \rightarrow a_n$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{\cos((n+1)\pi/3)}{(n+1)!} \times \frac{n!}{\cos(n\pi/3)} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{\cos((n+1)\pi/3)}{\cos(n\pi/3)} \times \frac{1}{n+1} \right|$$

$0 < \cos \leq 2$ for all $n \geq 1$.

$$= \lim_{n \rightarrow \infty} \left| \frac{\cos((n+1)\pi/3)}{\cos(n\pi/3)} \right| \times \lim_{n \rightarrow \infty} \left| \frac{1}{n+1} \right| = 0 < 1$$

//

By the Ratio Test, the given series is absolutely convergent. $\therefore$ converges //

$$(3) \sum_{n=1}^{\infty} \frac{n!}{n^n} \rightarrow a_n$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{(n+1)!}{(n+1)^{n+1}} \times \frac{n^n}{n!} \right| \\ &= \lim_{n \rightarrow \infty} \left| \left(\frac{n+1}{n} \right)^{-n} \right| \\ &= \lim_{n \rightarrow \infty} \left| \left(1 + \frac{1}{n} \right)^{-n} \right| \\ &= \frac{1}{e} < 1 \end{aligned}$$

By the Ratio Test, the given series is absolutely convergent. $\therefore \sum \frac{n!}{n^n}$ is convergent. //

12. Use any test to determine whether the series is absolutely convergent, conditionally convergent, or divergent.

$$(1) \sum_{n=1}^{\infty} \frac{n 5^{2n}}{10^{n+1}} \rightarrow a_n$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{(n+1) 5^{2n+2}}{10^{n+2}} \times \frac{10^{n+1}}{n 5^{2n}} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{n+1}{n} \times \frac{5}{2} \right| = \frac{5}{2} > 1 \end{aligned}$$

By the Ratio Test, the given series is divergent.

$$(2) \sum_{n=1}^{\infty} \frac{(-1)^n \arctan n}{n^2}$$

$$0 < \left| \frac{(-1)^n \arctan n}{n^2} \right| \leq \frac{\frac{\pi}{2}}{n^2} \sim \frac{\pi}{2} \sum_{n=1}^{\infty} \frac{1}{n^2} : p\text{-series with } p=2 \rightarrow \text{convergent.}$$

Thus, the given series is absolutely convergent. by DCT.

$$(3) \sum_{n=2}^{\infty} \frac{(-1)^n}{\sqrt{n} \ln n} \quad [\text{Hint : } \ln x < \sqrt{x}]$$

$$b_n = \frac{1}{\sqrt{n} \ln n}$$

$$\left| \frac{(-1)^n}{\sqrt{n} \ln n} \right| = \frac{1}{\sqrt{n} \ln n} > \frac{1}{n}$$

$$\sum_{n=2}^{\infty} \frac{1}{n} : p\text{-series with } p=1, \text{ divergent}$$

$$\text{So, } \left| \frac{(-1)^n}{\sqrt{n} \ln n} \right| \text{ is divergent.}$$

By AST, the given series is convergent. ① $b_{n+1} \leq b_n$
② $\lim_{n \rightarrow \infty} b_n = 0$

Thus, the given series is conditionally convergent.